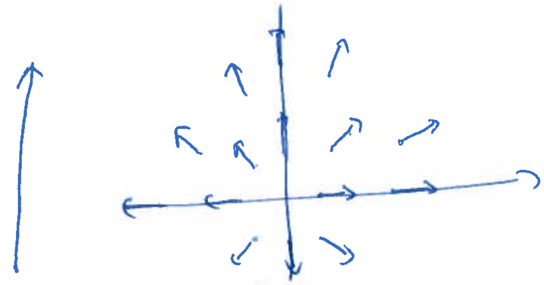
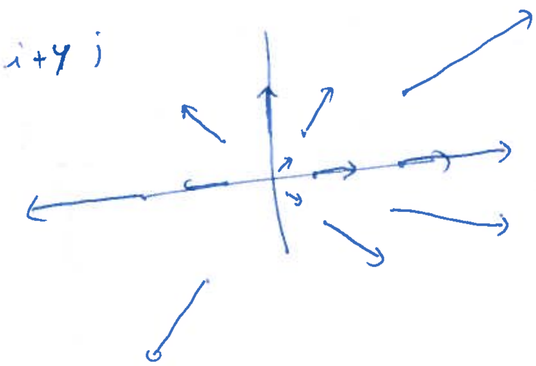
§14 Vector Calculus

Vector fields

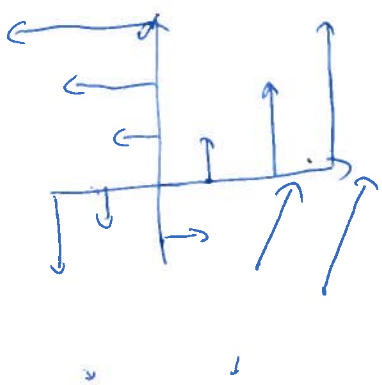
Ex $\quad F(x,y) = \dfrac{x\,i + y\,j}{\sqrt{x^2+y^2}}$ = unit vector in direction of (x,y)



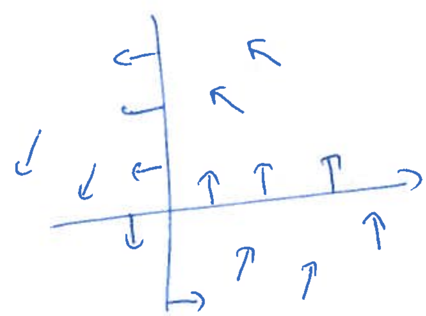
$F(x,y) = x\,i + y\,j$



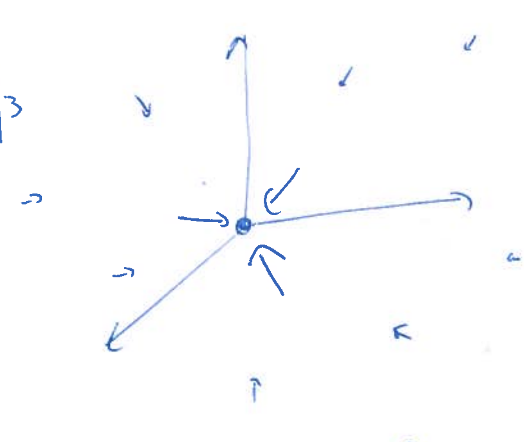
$F(x,y) = -y\,i + x\,j$
$= (-y, x)$



$F(x,y) = \dfrac{-y\,i + x\,j}{\sqrt{x^2+y^2}}$



Ex $\quad -\dfrac{GM}{p}\,\dfrac{\vec{r}}{\|r\|^3}$



Gradient of a function (scalar field)

$$\vec{F}(x,y,z) = \nabla f(x,y,z) = f_x \cdot \underline{i} + f_y \cdot \underline{j} + f_z \cdot \underline{k}$$

Eg $\quad f(x,y,z) = \dfrac{1}{(x^2+y^2+z^2)^{1/2}} = (x^2+y^2+z^2)^{-1/2}$

$$\nabla f = i\left(-\frac{1}{2}(x^2+y^2+z^2)^{-3/2} \cdot 2x\right) + j\left(-\frac{1}{2}(x^2+y^2+z^2)^{-3/2} \cdot 2y\right) + k\left(-\frac{1}{2}(x^2+y^2+z^2)^{-3/2} \cdot 2z\right)$$

$$= -\frac{1}{(x^2+y^2+z^2)^{3/2}}(xi+yj+zk) = -\frac{r}{\|r\|^3}$$

Divergence

$F = Mi + Nj + Pk \qquad \text{div } F = M_x + M_y + P_z$

$\text{div } F = \nabla \cdot F \qquad$ where $\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) \qquad F = (M, N, P)$

eg $\quad \text{div}\left(-\frac{r}{\|r\|^3}\right) = -\frac{1}{(x^2+y^2+z^2)^{3/2}} + \frac{3}{2}\frac{2x \cdot x}{(x^2+y^2+z^2)^{5/2}} + \frac{-1}{(x^2+y^2+z^2)^{3/2}} + \frac{3}{2}\frac{2y \cdot y}{(x^2+y^2+z^2)^{5/2}} \frac{-1}{(x^2+y^2+z^2)^{3/2}}$

$$+ \frac{3}{2}\frac{z \cdot z}{(x^2+y^2+z^2)^{5/2}}$$

$$= \frac{1}{\|r\|^5}\left(-3(x^2+y^2+z^2) + 3x^2 + 3y^2 + 3z^2\right) = 0$$

Curl

$$\text{Curl } F = (P_y - N_z)i + (M_z - P_x)j + (N_x - M_y)k$$

$$= \nabla \times F = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ M & N & P \end{vmatrix}$$

eg $\quad \nabla \times (xyi + zj + yxk)$

$$= \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xy & z & yx \end{vmatrix} = i(x-1) - j(y-0) + k(x-0) = (x-1, -y, x)$$

Eg $F = \nabla f$ then $\nabla \times F = \nabla \times \nabla f = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ f_x & f_y & f_z \end{vmatrix}$

$= i(f_{zy} - f_{yz}) - j(f_{zx} - f_{xz}) + k(f_{xy} - f_{yx}) = 0$ if second deriv. are cont.

Eg $\nabla \times \left(-\frac{\mathbf{r}}{\|r\|^3}\right) = \nabla \times \nabla\left(\frac{1}{\|r\|}\right) = 0$

Eg $\nabla \cdot (\nabla f \times \nabla g) = \begin{vmatrix} \partial/\partial x & \partial/\partial y & \partial/\partial z \\ f_x & f_y & f_z \\ g_x & g_y & g_z \end{vmatrix} = (f_y g_z - f_z g_y)_x - (f_x g_z - f_z g_x)_y + (f_x g_y - f_y g_x)_z$

$= f_{yx} g_z + f_y g_{zx} - f_{zx} g_y - f_z g_{yx} - f_{xy} g_z - f_x g_{zy} - f_{zy} g_x - f_z g_{xy} + f_{xz} g_y + f_x g_{yz} - f_{yz} g_x - f_y g_{xz}$

$= 0$

Ex Show that $\operatorname{div}(F \times G) = G \cdot \operatorname{curl} F - F \operatorname{curl} G$

Ex Sketch the vector fields a) $F(x,y) = (-x, 2y)$ b) $F(x,y) = 3x\,i + y\,j$

c) $F(x,y,z) = -z \cdot k$

Ex Compute ∇F a) $F = x^2 - 3xy + 2z$ b) $f(x,y,z) = \ln|xyz|$

c) $f(x,y,z) = y^2 e^{-2z}$

Ex Compute $\operatorname{div} F$ & $\operatorname{curl} F$

a) $F(x,y,z) = x^2 i + y^2 j + z^2 k$ b) $F(x,y,z) = yz\,\underline{i} + xz\,\underline{j} + xy\,\underline{k}$

Ex Show that $\operatorname{curl}(f\vec{F}) = f \operatorname{curl}(F) + (\operatorname{grad} f) \times F$